

Module 2 – Logistic Regression

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Introduction

- In the [Linear regression module](#), we explored how to construct a model to make continuous numerical predictions, such as the fuel efficiency of a car.
- Many real problems require **decisions**.

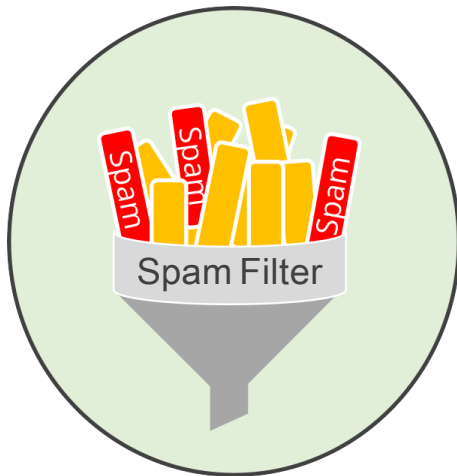
Use Cases



Loan sanction



Customer segments



Spam filtering

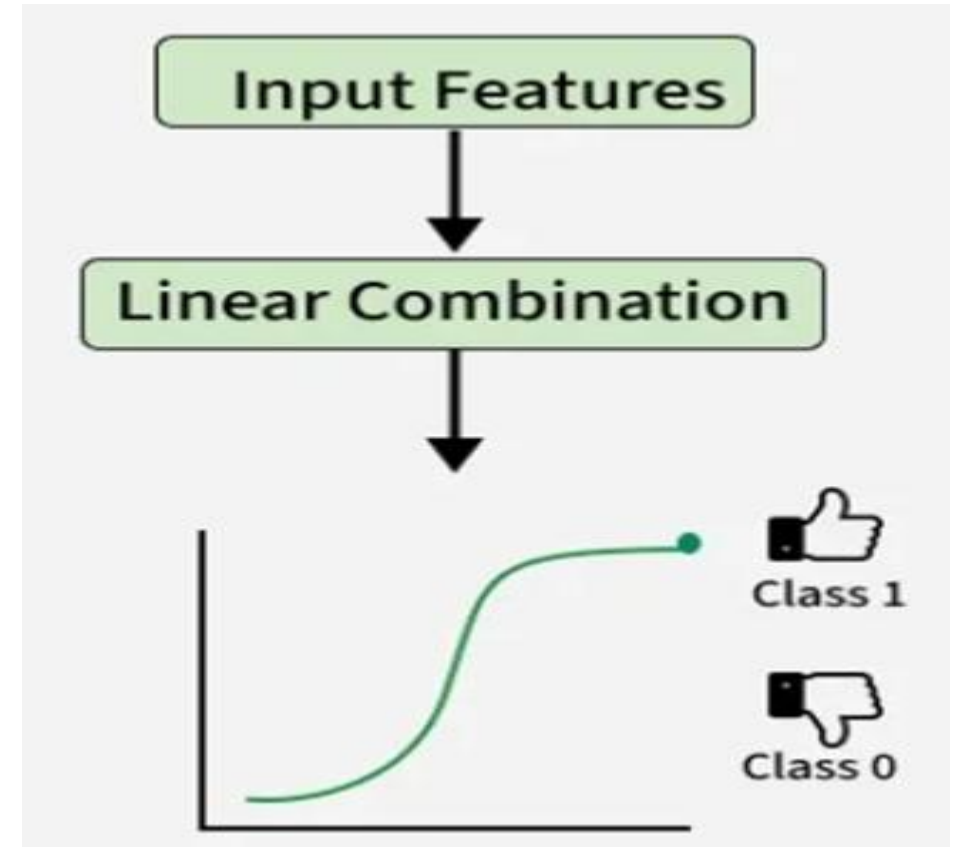


Exam results – Pass/Fail

Logistic Regression

- Logistic Regression is widely used to predict binary outcomes for a given set of independent variables.
- The dependent variable's outcome is discrete such as

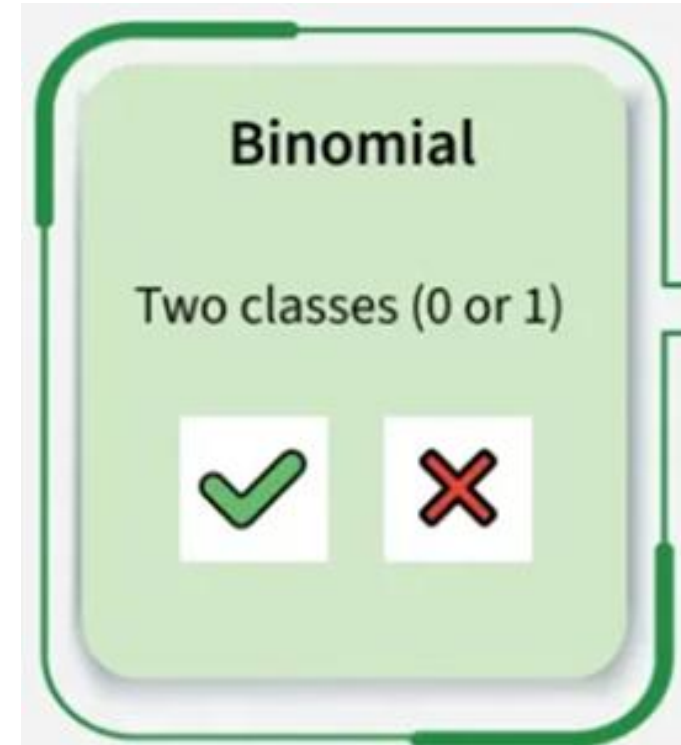
$$y \in \{0, 1\}$$



Types of Logistic Regression

Binary (Binomial)

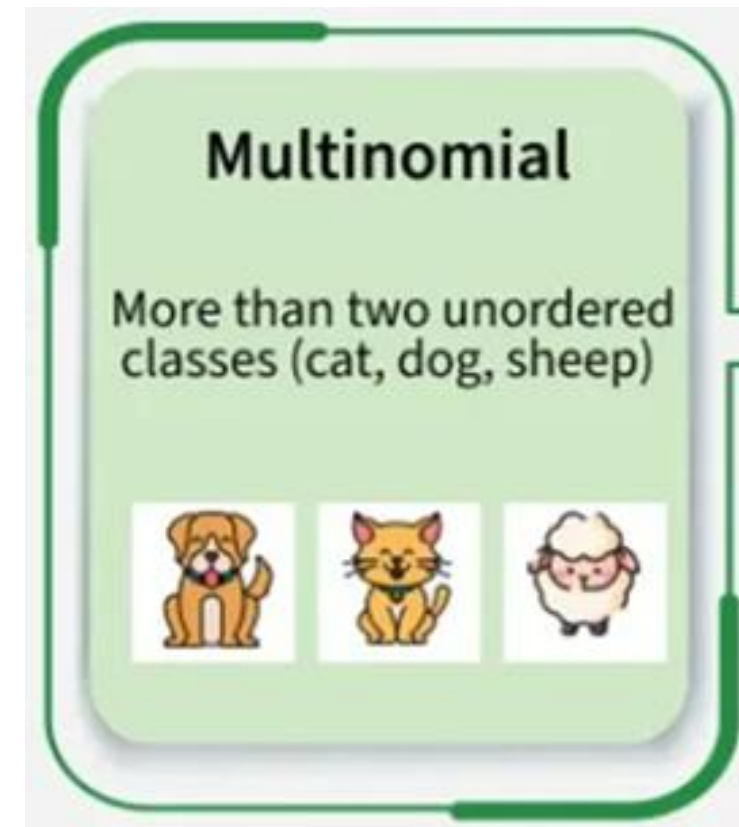
- The dependent variable has **only two possible categories**.
- These categories are typically coded as **0 and 1**.



Types of Logistic Regression (Contd.)

Multinomial Classification

- The dependent variable has **three or more categories**.
- These categories are **unordered** and have **no inherent ranking or quantitative meaning**.



Types of Logistic Regression (Contd.)

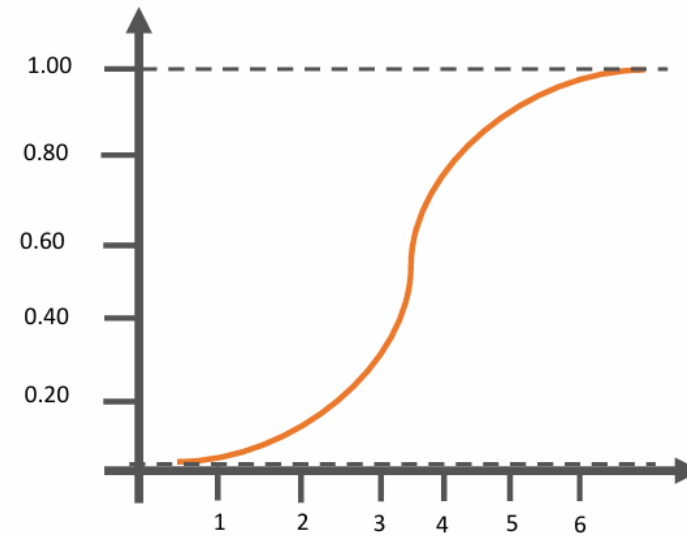
Ordinal Classification

- The dependent variable has **three or more categories**.
- These categories are **ordered** and carry **quantitative significance** (a natural ranking).

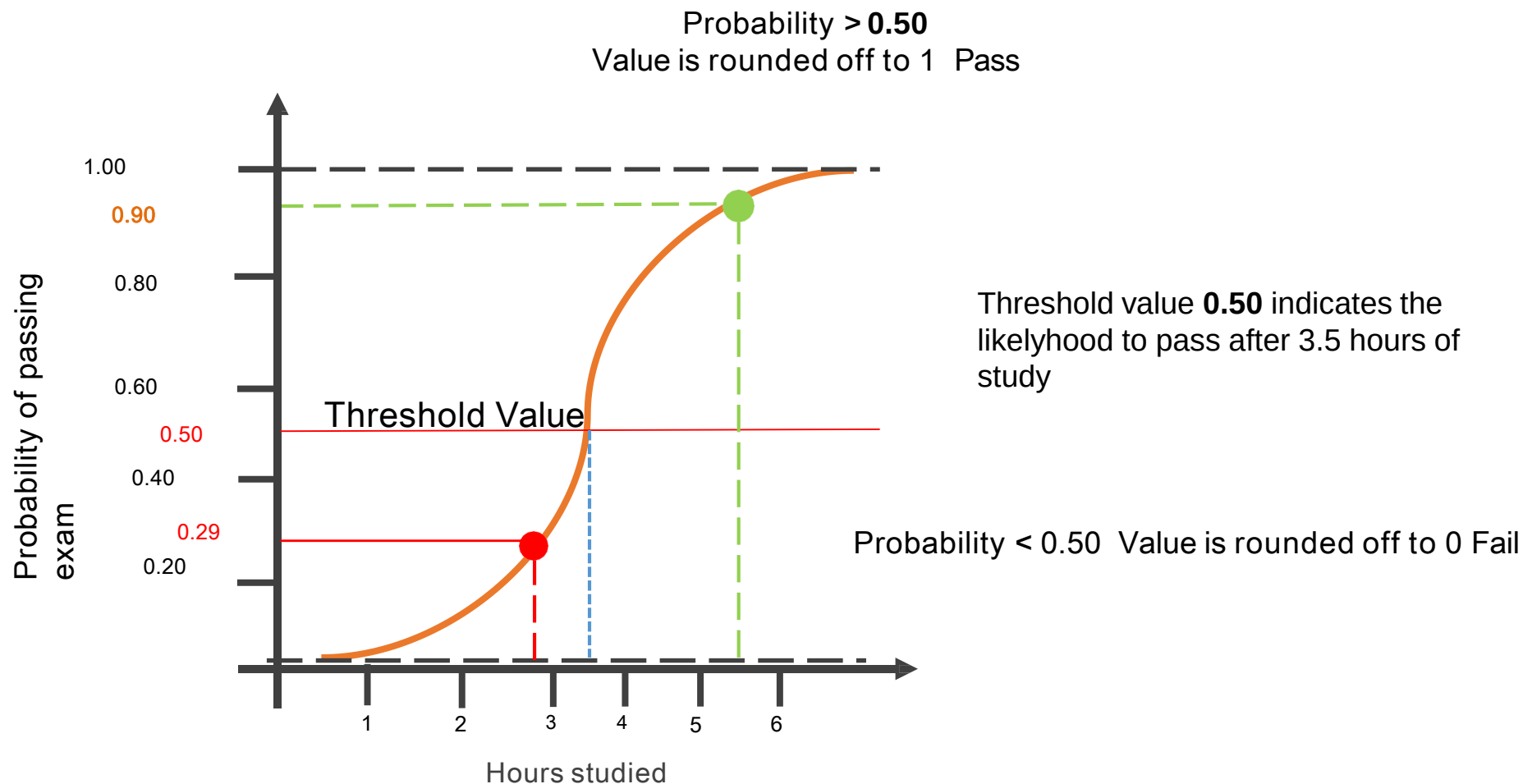


Sigmoid Function

- The probability distribution of output y is restricted to 1 or 0. This is called as **sigmoid probability (σ)**
- Produces an **S-shaped curve** (logistic curve).
- Common threshold: **0.5**
 - Output $\geq 0.5 \rightarrow$ **Class 1**
 - Output $< 0.5 \rightarrow$ **Class 0**



Sigmoid Function (Contd.)



Assumptions of Logistic Regression

- **Binary Outcome** →
 - Dependent variable must be dichotomous (two categories).
- **Independence of Observations** →
 - Each data point should be independent.
- **No Multicollinearity** →
 - Predictors should not be highly correlated.

Assumptions of Logistic Regression

- **Linearity with Log-Odds →**
 - Independent variables must have a linear relationship with the log-odds of the outcome.
- **No Extreme Outliers →**
 - Outliers should be minimized to avoid distortion.
- **Large Sample Size →**
 - Requires sufficient data for stable and reliable estimates.

Logistic Regression Terminologies

- **Odds** → Ratio of the probability of an event occurring to the probability of it not occurring.
- **Log-Odds (Logit)** → Natural logarithm of the odds; modelled as a linear combination of predictors and the intercept.
- **Coefficient** → Parameters estimated by the model; indicate the strength and direction of the effect of independent variables on the dependent variable.
- **Intercept** → Constant term; represents the log-odds when all predictors are zero.
- **Maximum Likelihood Estimation (MLE)** → Method used to estimate coefficients by maximizing the likelihood of observing the given data.

How Does Logistic Regression Work

Input Representation

Suppose we have a dataset with **n features** and **m training examples**, represented by the feature matrix:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}$$

The dependent variable Y is **binary**, taking only two possible values:

$$Y = \begin{cases} 0 & \text{if Class 1} \\ 1 & \text{if Class 2} \end{cases}$$

How Does Logistic Regression Work (Contd.)

Logistic regression first computes a **linear combination** of the input features:

$$z = \sum_{i=1}^n w_i x_i + b$$

- Where:
 - x_i is the i -th input feature
 - w_i are the weights (coefficients)
 - b is the bias (intercept)

This can be written compactly as a dot product: $z = w \cdot X + b$

At this stage, z is a continuous value ranging from $-\infty$ to $+\infty$, just like linear regression output.

How Does Logistic Regression Work (Contd.)

Sigmoid Function

To convert this continuous value into a probability, logistic regression applies the **sigmoid function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

Properties of the Sigmoid Function:

- $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$
- $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$
- Output is always bounded between 0 and 1

Thus the predicted probabilities are:

$$P(y = 1) = \sigma(z)$$

$$P(y = 0) = 1 - \sigma(z)$$

Example

- The dataset of pass or fail in an exam of 5 students is given in the table. Use logistic regression as classifier to answer the following questions.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

- Calculate the probability of pass for the student who studied 33 hours.
- At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 * \text{hours}$

Example

1. Calculate the probability of pass for the student who studied 33 hours.

$$p = 1 / (1 + e^{(-z)})$$

$$z = -64 + 2 * 33 = -64 + 66 = 2$$

$$p = 1 / (1 + e^{-2}) = 0.88$$

That is, if student studies 33 hours, then there is 88% chance that the student will pass the exam

Hours Study	Pass (1) / Fail (0)
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39	1

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 * \text{hours}$

Example

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Given $p = \frac{1}{1 + e^{-z}}$ $p=0.95$

Substitute probability into sigmoid equation

$$0.95(1 + e^{-z}) = 1$$

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 * \text{hours}$

Hours Study	Pass (1) / Fail (0)
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Example

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$0.95 + 0.95e^{-z} = 1$$

$$e^{-z} = \frac{0.05}{0.95} = 0.0526$$

$$0.95e^{-z} = 1 - 0.95$$

$$0.95e^{-z} = 0.05$$

$$\ln(e^{-z}) = \ln(0.0526)$$

$$z = -64 + 2(\text{hours})$$

$$z = 2.94$$

$$\text{hours} = 33.47$$

Divide both sides by 0.95:

$$e^{-z} = \frac{0.05}{0.95} = 0.0526$$

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 * \text{hours}$

Overview

- [Click Here](#)

Cost Function – Can we use MSE?

- A **cost function** (also known as a loss function) measures the error between the actual values (ground truth) and the predicted values by a model.
- The goal of most machine learning algorithms is to minimize this cost, which leads to better predictions.

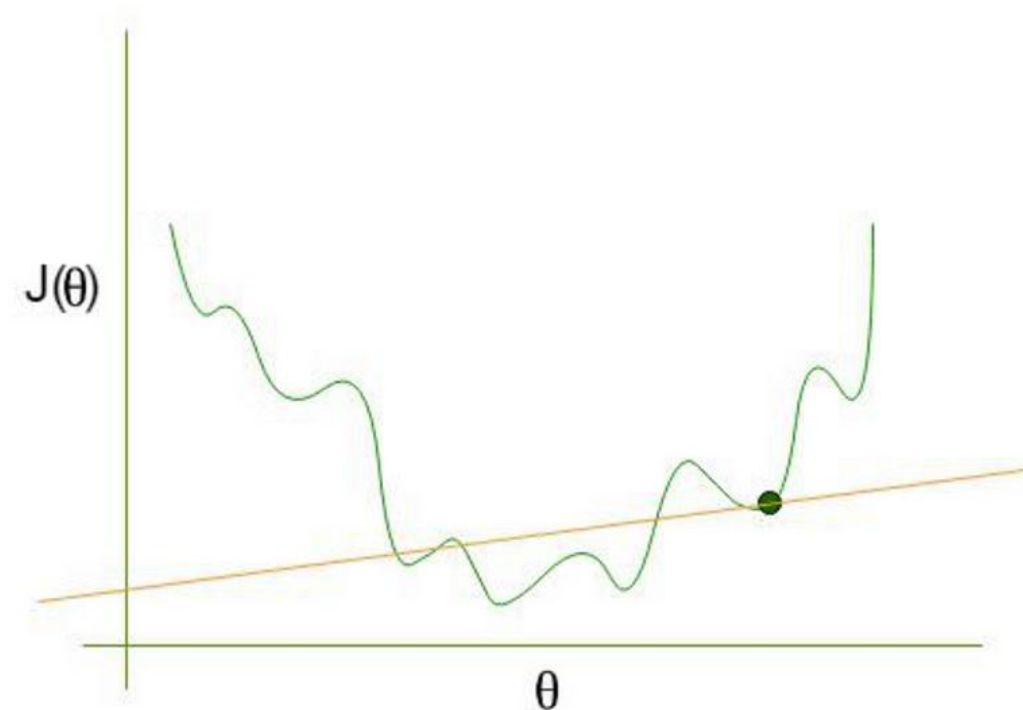
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \tilde{y}_i)^2$$

Why MSE is not Suitable?

1. Nonlinearity and Non-Convex Cost Function

The sigmoid function introduces **nonlinearity** into the model.

When this nonlinear sigmoid output is substituted into the MSE cost function, the resulting error surface becomes **non-convex**.



Why MSE is not Suitable?

2. Ineffective Learning Due to Squared Errors

- Mean Squared Error squares the difference between the predicted probability and the actual class label (0 or 1).
- Logistic Regression outputs probabilities in the range **[0, 1]**
- Squaring these small differences can make gradients **very small**
- This leads to **slow learning** and poor weight updates
- As a result, the model struggles to learn correct decision boundaries.

Log Loss (Cross-Entropy)

- Logistic Regression uses **Log Loss**, also known as **Cross-Entropy Loss**:

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m [y_i \log(h(x_i)) + (1 - y_i) \log(1 - h(x_i))]$$

Where

- m is the number of training examples
- y_i is the true class label for the i -th example (either 0 or 1).
- $h\theta(x_i)$ is the predicted probability for the i -th example, as calculated by the logistic regression model.
- θ is the model parameters

Logistic Regression Equation and Odds

- **Odds** The odds represent the ratio of the probability of an event occurring to the probability of it not occurring:

$$\text{ODD} = p(x)/(1-P(x))$$

Unlike probability (which ranges from 0 to 1), odds range from 0 to ∞ .

- **Log-Odds (Logit)** The log-odds are the natural logarithm of the odds. In logistic regression, the log-odds are expressed as a linear combination of the independent variables and the intercept.

Logistic Regression Equation and Odds

$$\log \left[\frac{p(x)}{1 - p(x)} \right] = z$$

$$\log \left[\frac{p(x)}{1 - p(x)} \right] = w \cdot X + b$$

$$\frac{p(x)}{1 - p(x)} = e^{w \cdot X + b} \quad \dots \text{Exponentiate both sides}$$

$$p(x) = e^{w \cdot X + b} \cdot (1 - p(x))$$

$$p(x) = e^{w \cdot X + b} - e^{w \cdot X + b} \cdot p(x)$$

$$p(x) + e^{w \cdot X + b} \cdot p(x) = e^{w \cdot X + b}$$

$$p(x)(1 + e^{w \cdot X + b}) = e^{w \cdot X + b}$$

$$p(x) = \frac{e^{w \cdot X + b}}{1 + e^{w \cdot X + b}}$$

$$p(x) = \frac{e^{w \cdot X + b}}{1 + e^{w \cdot X + b}} = \frac{1}{1 + e^{-(w \cdot X + b)}}$$

Logistic Regression Equation and Odds

- **Logistic Regression Model Equation**

Substituting $z=w \cdot X+b$ into the sigmoid function:

$$p(X; w, b) = \frac{1}{1 + e^{-(w \cdot X + b)}}$$

This equation gives the **probability that the input belongs to Class 1**.

Maximum Likelihood Estimation (MLE)

To estimate the parameters w and b , logistic regression uses **Maximum Likelihood Estimation (MLE)**.

- For a single data point:
 - If $y=1$: probability = $p(x)$
 - If $y=0$: probability = $1-p(x)$

The **likelihood function** for all training samples is:

$$L(b, w) = \prod_{i=1}^n p(x_i)^{y_i} (1 - p(x_i))^{1-y_i}$$

Log-Likelihood Function

- Taking the natural logarithm simplifies optimization:

$$\log L(b, w) = \sum_{i=1}^n [y_i \log p(x_i) + (1 - y_i) \log(1 - p(x_i))]$$

- This expression is known as the **log-likelihood function**.
- Maximizing this function is equivalent to **minimizing the log-loss (cross-entropy loss)**.

Gradient of the Log-Likelihood

- The gradient with respect to each weight w_j is:

$$\frac{\partial \log L}{\partial w_j} = \sum_{i=1}^n (y_i - p(x_i; w, b)) x_{ij}$$

- This update rule adjusts the weights in the direction that **reduces the difference between actual labels and predicted probabilities.**

Linear Regression Vs Logistic Regression

Linear Regression

- Used to solve regression problems.
- The response variables are continuous in nature
- It helps estimate the dependent variable when there is a change in the independent variable.
- It is a straight line.

Logistic Regression

- Used to solve classification problems
- The response variable is categorical in nature
- It helps calculate the possibility of a particular event taking place.
- An S-curved line.

Example

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Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 * \text{hours}$

Example

1. Calculate the probability of pass for the student who studied 33 hours.

- The optimizer suggests the **log-odds model**:

$$\log \left(\frac{p}{1-p} \right) = -64 + 2 \times (\text{Hours})$$

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 \times \text{hours}$

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$$\log \left(\frac{p}{1-p} \right) = -64 + 2 \times (\text{Hours})$$

$$\begin{aligned}
 \log \left(\frac{p}{1-p} \right) &= -64 + 2(33) \\
 &= -64 + 66 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 \frac{p}{1-p} &= e^2 \\
 e^2 &\approx 7.389
 \end{aligned}$$

$$\begin{aligned}
 p &= \frac{7.389}{1 + 7.389} \\
 p &= \frac{7.389}{8.389} \\
 p &\approx 0.88
 \end{aligned}$$

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 \times \text{hours}$

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Example

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

- **Set Probability Condition** $p > 0.95$
- Convert Probability to Odds
- Convert Odds to Log-Odds
- Substitute into Logistic Model

$$\frac{p}{1-p} > \frac{0.95}{0.05}$$

$$\frac{p}{1-p} > 19$$

$$\log\left(\frac{p}{1-p}\right) > \log(19)$$

$$\log(19) \approx 2.944$$

$$-64 + 2h > 2.944$$

$$2h > 66.944$$

$$h > 33.472$$

Assume the model suggested by the optimizer for odds of passing the course is, $\log(\text{odds}) = -64 + 2 \times \text{hours}$

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Example

Construct a logistic regression model with two predictors for the riding mower example with $\beta_0 = -25.9382, \beta_1 = 0.1109, \beta_2 = 0.9638$, where β_1 and β_2 are for the "Income" and "Lot_Size" variables, respectively.

Cust	Income	Lot_Size	Ownership
1	60.0	18.4	Owner
2	64.8	21.6	Owner
3	84.0	17.6	Nonowner
4	59.4	16.0	Nonowner
5	108.0	17.6	Owner
6	75.0	19.6	Nonowner

Example (Contd.)

General Model:

$$P(y=1|x) = 1 / (1 + e^{(-z)})$$

- So

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

- Substituting Given Values

$$z = -25.9382 + 0.1109(\text{Income}) + 0.9638(\text{Lot Size})$$

Example (Contd.) Customer 1 Data

Customer	Income	Lot Size	Ownership
1	60.0	18.4	Owner

- Compute Linear Combination (z)

$$z = -25.9382 + 0.1109(60) + 0.9638(18.4)$$

$$z = -25.9382 + 6.654 + 17.734$$

$$z = -1.5502$$

- Apply Sigmoid Function

- $P(y=1) = 1 / (1 + e^{(1.5502)})$

- ≈ 0.175

Example (Contd.)

- Classification

Threshold = 0.5

$$0.175 < 0.5$$

- Predicted Class = Nonowner



Thank You

- IMP
- <https://www.appliedaicourse.com/blog/logistic-regression-in-machine-learning/>
- <https://www.ibm.com/think/topics/logistic-regression>